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| Name:                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  MANIPAL UNIVERSITY<br>JAIPUR                                       |             |
| Enrolment No:                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| Odd Semester Make Up Examination, Dec 2025<br>FoSTA, School of Computer Science & Engineering<br>Department of Artificial Intelligence & Machine Learning<br>B.Tech. CSE (AIML) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| Course Code: AIM3144                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Course: High Performance Computing                                                                                                                  | Semester: V |
| Time: 03 hrs.                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Max. Marks: 80                                                                                                                                      |             |
| Instructions: All questions are compulsory.<br>Missing data, if any, may be assumed suitably.<br>Calculator is allowed.                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| SECTION A (Memory Based Questions)                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| S.No.                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mark                                                                                                                                                | CO          |
| Q A1                                                                                                                                                                            | Compare between shared memory and message passing models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2*5= Max Marks 10                                                                                                                                   | CO1         |
| Q A2                                                                                                                                                                            | How does the selection of an interconnection topology determine communication cost and performance in parallel processors? Explain with suitable reasoning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                     | CO4         |
| Q A3                                                                                                                                                                            | Why the MIPS architecture chooses base-displacement addressing over indirect addressing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                     | CO3         |
| Q A4                                                                                                                                                                            | A pipeline takes 17 cycles to execute 8 instructions. How many stages (k) does the pipeline have?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                     | CO4         |
| Q A5                                                                                                                                                                            | Choose the correct answer in following question.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                     | CO4         |
|                                                                                                                                                                                 | (i) If pipeline registers have no delay, the total time to execute N instructions in a 5-stage pipeline is:<br>A. 5N<br>B. N + 5<br>C. (N + 4) × (max stage delay)<br>D. N × (max stage delay)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                     |             |
|                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (ii) In MPIS-1, coherence is maintained using:<br>A. Directory protocol<br>B. Bus snooping<br>C. Token-based approach<br>D. Distributed directories |             |
| SECTION B (Concept Based Questions)                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| Q B1                                                                                                                                                                            | The following MIPS instructions operate on registers and memory:<br>ADD R1, R2, R3 # Instruction 1<br>LW R4, 8(R2) # Instruction 2<br>ADDI R5, R4, 5 # Instruction 3<br>SW R6, 16(R3) # Instruction 4<br>a) Demonstrate your understanding of MIPS addressing by identifying the addressing mode used in each of the above four instructions.<br>b) R2 = 0x10010000, apply the addressing rule to compute the effective address used in Instruction 2.<br>c) Given R3 = 0x10010010, compute the effective address used in Instruction 4 by applying your knowledge of base-displacement addressing.<br>d) If the value in R4 = 0x20, use the ADDI instruction to determine the resulting value placed in register R5 after Instruction 3 executes. | 5*6= Max Marks 30                                                                                                                                   | CO2         |
| Q B2                                                                                                                                                                            | Given the MIPS instructions:<br>SUB R1, R2, R3 # Instruction 1<br>LW R4, 12(R2) # Instruction 2<br>ADDI R5, R4, 10 # Instruction 3<br>SW R6, 20(R3) # Instruction 4<br>a) If R2 = 0x 20010000, compute the effective address for Instruction 2?<br>b) If R3 = 0x 20010020, compute the effective address for Instruction 4?<br>c) If R4 = 0x 15, identify the value in R5 after Instruction 3 executes?                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                     | CO1         |
| Q B3                                                                                                                                                                            | Given the decimal number 0.25, demonstrate your understanding of the IEEE 754 single-precision format by performing the following steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                     | CO2         |

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|                                                   | a) Identify the sign bit, exponent, and mantissa for 0.25 in 32-bit floating-point representation.<br>b) Combine these fields to construct the complete 32-bit IEEE 754 representation of 0.25.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |     |
| Q B4                                              | Given the following MIPS instructions:<br><div>ADD R1, R2, R3 # Instruction 1<br/>LW R4, 8(R1) # Instruction 2<br/>SW R5, 12(R1) # Instruction 3<br/>JR R6 # Instruction 4</div> a) Calculate the total cycles needed and describe the role of each stage<br>b) Determine the number of cycles and explain the function of each stage, including memory access.<br>c) Compute the total cycles and describe each stage, highlighting the memory write operation.<br>d) Compute the cycles required and explain the role of each stage in executing this instruction                                                                                                                     |               | CO5 |
| Q B5                                              | Given the following scenario: <ul style="list-style-type: none"><li>• Processor P1 and P2 have local caches that hold a copy of a shared variable A.</li><li>• Initially, A = 100 in memory.</li><li>• P1 writes A = 200 to its cache.</li><li>• P2 attempts to read A after P1's write operation.</li></ul> a) Identify the cache coherence problem that may occur in this scenario.<br>b) Illustrate how the write-once protocol addresses this cache coherence problem, describing the mechanism step by step.<br>c) Analyze the consequences if Processor P2 does not monitor (snoop) the bus in the system. Explain what effect this would have on data consistency and coherence. |               | CO3 |
| SECTION-C (Analytical Based Questions)            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |     |
| Q C1                                              | A non-pipelined system takes 50 ns to complete a single task. The same task can be executed on a six-stage pipeline, where each stage operates with a clock cycle of 10 ns. Analyze how the total execution time for processing 500 tasks differ between the non-pipelined and pipelined systems, and evaluate the approximate speedup ratio achieved by using the pipeline.                                                                                                                                                                                                                                                                                                            | 2*10=20 Marks | CO4 |
| Q C2                                              | A non-pipelined processor operates at a clock frequency of 3 GHz and requires 5 cycles to execute each instruction. The processor is then upgraded to a 5-stage pipeline, but the clock speed decreases to 2.4 GHz due to the overhead introduced by pipelining. Assuming ideal conditions with no pipeline stalls, analyze the execution time per instruction for both the non-pipelined and pipelined designs, and evaluate the performance improvement by calculating the speedup achieved. Discuss whether the pipeline upgrade provides a meaningful performance gain despite the reduction in clock frequency.                                                                    |               | CO4 |
| SECTION-D (Case Study/Application based Question) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |     |
| Q D1                                              | Design a problem that requires calculating the total execution time for a 5-stage pipeline with given stage delays of 50 ns, 60 ns, 40 ns, 55 ns, and 45 ns, where each interstage register adds an additional delay of 5 ns. Assume ideal conditions with no stalls or hazards and a constant clocking rate. Using this setup, determine the total time required to process 300 tasks. Structure your solution to show how the stage delays, register delays, and the number of tasks are combined to compute the final execution time.                                                                                                                                                | 2*10=20 Marks | CO5 |
| Q D2                                              | Consider the following scenario in an MPI-based system:<br>➤ Process P0 wants to broadcast a message to all other processes.<br>a) Develop a clear example showing how a message can be broadcast from a single process to all other processes in MPI. Identify and explain the MPI function that performs this broadcast.<br>b) Develop a comparative example that illustrates the difference between point-to-point communication and collective communication in MPI. Explain each type and provide a simple example demonstrating how data is exchanged in both cases.                                                                                                              |               | CO4 |